

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 30

Code of Conduct

I N. Sai Steenu / 192365001 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N. Sai Steenu

Assessment Tasks

Class Test

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.
2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.
3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.
4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.
5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.
6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.
7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.
8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.

9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.
10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points.	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

1) Unified Big Data Architecture for multinational Retail.

A multinational retail organization may have customer data stored separately in websites, mobile apps, physical stores and social media. This fragmentation makes it difficult to obtain a complete view of customer

behaviour

A suitable architecture is:

Data sources → Data Ingestion → Data Lake → processing → Unified Customer profile → Analytics/ML → Recommendation

- Data sources: online purchases, pop transactions, mobile apps, websites, loyalty cards and records.
- Data Ingestion: Apache Kafka can collect real-time transaction and clickstream data.
- Data storage: Hadoop HDFS or cloud object can store large volume of structured data.
- processing: Apache spark can perform batch and processing.
- Data Integration: customer ID's, email ID's, and loyalty IDs can be matched to create customer profiles
- Analytics: machine learning can identify customer ID's purchase, patterns and customer segments.

2) Distributed computing for scientific Research

scientific research generates huge datasets from satellites, medical experiments, simulations, genomics and particle physics. processing such data using a centralized system can be slow and expensive.

Centralized system:-

- one main computer processes the entire dataset.
- Limited processing capacity.
- single point of failure.

Distributed system:-

- Data and computation are distributed across multiple nodes.
- Tasks execute in parallel.
- processing time is reduced
- Additional time can be added easily

Technologies such as Hadoop map Reduce and Apache spark support distributed processing.

Therefore, parallel process improves speed, scalability making distributed computing.

3) Big Data Fraud Detection Framework

A digital payment platform receives millions of transactions from different locations and devices. Traditional fraud detection may not identify fraudulent transactions quickly enough.

A suitable Framework is:

Transactions $\rightarrow$ Kafka $\rightarrow$ stream processing $\rightarrow$ Feature engineering
ML model $\rightarrow$ fraud store $\rightarrow$ Alert/Block.

- Data Sources: Transaction amount, location, device time, customer history and previous fraud records.
- Kafka: collects payment transactions in real time.
- spark streaming: processes transaction (or) history.
- Feature engineering: calculates unusual transaction, location changes transaction frequency.
- Fraud store: Each transaction receives a probability.
- Action: High-risk transactions can be blocked or sent for manual verification.

4) Multimedia data in Big Data

Multimedia data includes videos, audio files and images. These datasets create several challenges because they are large, unstructured and expensive to process.

Major challenges:

- Very large file sizes
- Unstructured data formats
- High storage requirement.
- Complex processing and analysis.

Suitable Technologies:

Storage:

- Hadoop HDFS
- Cloud object storage such as Amazon S3
- Distributed file systems.
- No SQL datasets for metadata.

Processing:

- Apache Spark
- Hadoop Map Reduce
- GPU-based processing.

Analysis:

- Computer vision for images and videos
- Speech recognition for audio.
- Deep learning.

5)

Disaster Recovery and Business continuity

A cloud based big data platform must continue even when hardware, software, network or cloud-region failures occur.

A suitable strategy includes:

Primary system → Replication → Backup/secondary system → Automated failover.

Key components.

1. Redundancy: maintain multiple servers and systems.
2. Data Replication: Replicate important data sets across multiple nodes or regions.
3. Regular backups: create scheduled backups and maintain recovery copies.
4. Disaster recovery site: maintain a secondary environment.
5. Automated failover: Automatically redirect applications to healthy systems.
6. Monitoring: continuously monitor system health.
7. Recovery testing: Regularly test restoration and failover procedures.

6) Data Quality and Data Governance:

Poor-quality data can seriously affect the reliability of big data analytics.

Common data quality problems:

- Inconsistency: Different systems use different formats or values.
- Duplication: The same customer or transaction appears multiple times.
- missing values: Important fields may be unavailable.
- Incorrect values: Data may contain errors.

Data Governance strategies:

- Establish data quality standards
- use master data management
- Remove Duplicate records
- validate data during ingestion
- Handle missing value using appropriate techniques
- maintain data dictionaries and metadata.
- Define data ownership and responsibilities
- continuously monitor data quality.
- Implement access control and auditing.

7) How does Apache NiFi help?

→ NiFi helps move and integrate data between different systems.

→ It can collect, filter, transform and route data.

→ It also provides data tracking and monitoring.

8) Failure points in Hadoop:

→ common failures include data nodes, disks, networks and task failures

→ monitoring helps identify problems quickly.

Limitations:-

- Data security and privacy risks

- Dependence on cloud providers

- Network dependency

- unexpected cloud costs

- vendor lock-in

- compliance challenges.

Therefore, cloud computing provides excellent scalability and flexibility, but organizations must carefully manage security costs and compliance.

9) Real-time Transportation and logistics Analytics.

A suitable architecture is

GPS/vehicle sensors → Kafka → streaming processing → Analytics
→ Dash board / Alerts

Applications :-

Route optimization:

Real time traffic and GPS data can identify route

Fuel efficiency:

Vehicle speed, engine performance, distance and driving patterns

predictive maintenance:

Sensors data can identify possible vehicle failures.

10) Ethical and legal challenges of Big Data

Big data provides valuable insights but creates significant ethical and legal concerns.

Ethical challenges:

- Loss of user privacy
- Extensive data collection

Legal challenges:

Organizations must comply with applicable data protection and privacy regulations.

Recommended policies:

1. collect only necessary data
2. obtain informed user consent
3. Encrypt sensitive information.